

[AI-02] AI AUDIT REPORT

Course: Software Testing
Exercise ID: HW01-AI
Student Name: Lâm Hữu Khánh
Student ID (MSSV): 23127205

1. AI Audit Report ([AI-02])

(One batch generated by a single prompt counts as one artifact/entry)

Artifact 1: Scaffold Template

- **(1) Prompt + tool:**
 - **AI Tool Name:** Claude Sonnet 4.6
 - **Timestamp:** 09:00 02/06/2026
 - **Prompt:**

I have a software testing homework assignment called HW01-AI. It has 3 requirements:
Requirement 1: Find 10 QA/QC job postings (min 3 must require AI/LLM skills), with screenshots showing my logged-in account.
Requirement 2: Find 20 software defects from 2022-2026 (min 5 AI/LLM related). For each: source link, description, severity, consequences, solution, and 1 AI hallucination/bias instance.
Requirement 3: Choose a physical household device, design 15 test cases (with 3 edge cases AI missed), report 5 defects on GitHub, and record 5 YouTube execution videos.
The report must also include: AI Collaboration Protocol (Mindmap correction, AI Audit Report, AI Critique, Mandatory Disclosure) and a Self-Assessment Rubric.
Please generate a complete, clean Markdown report template (with all headings, subheadings, and placeholder text) for this homework so I can fill it in.

- **(2) AI output (full raw output — key structure preserved verbatim):**

```
# REPORT: HOMEWORK 01 (HW01-AI)

**Course:** Software Testing

## STUDENT INFORMATION

- **Full Name:** [Your Name]
- **Student ID:** [MSSV]
```

```
- **GitHub Repository Link:** [Link]

## REQUIREMENT 1: QA/QC JOB MARKET 2026+

### Summary Table of 10 Job Postings

| No. | Job Position | Company | AI Skills Required? | Salary | Job Link |
| ... | ... | ... | ... | ... | ... |

### Detailed Job Postings

#### 1. Job Posting 1: [Position] - [Company]

- **Job Link:** [URL]
- **Screenshot:** [image]
- **Job Description:** [...]
- **Required Skills:** [...]
- **Salary:** [...]
- **AI Impact Analysis:** [1-2 sentences]

## REQUIREMENT 2: 20 SOFTWARE DEFECTS 2022-2026

### DEFECT #1

#### 1. Defect Name & Year

#### 2. Source Link

#### 3. Description

#### 4. Severity

#### 5. Consequences

#### 6. Solution

#### 7. AI Bias / Hallucination Instance

| Category | Content |
| AI Tool Used | ... |
| AI Explanation Error | ... |
| Evidence | ... |

## REQUIREMENT 3: TEST CASES FOR ONE PHYSICAL PRODUCT

### 1. Device Under Test (DUT) Declaration

### 2. List of 15 Test Cases

| Test ID | Objective | Input | Steps | Expected | Actual | Verdict |

### 3. Analysis of 3 Edge Cases Missed by AI

### 4. Physical Defects Reporting (GitHub Issues)
```

```
### 5. List of Execution Videos (YouTube Unlisted)

## AI COLLABORATION PROTOCOL

### 1. QA/QC Role Mindmap Correction

### 2. AI Audit Report ([AI-02])

### 3. AI Critique (200-300 words)

### 4. Mandatory Disclosure

## Self-Assessment Rubric

| No. | Criteria | Max Grade | Self-Assessed Grade | Notes |
```

- (3) **Verdict: INCOMPLETE**
- (4) **Reasoning:** Per ISTQB CTFL Chapter 5 (Test Management) and the assignment's "Quality over Completion" principle, documentation templates must reflect the exact deliverable structure. The AI scaffold was missing the mandatory 7-section defect format, omitted the "Dated Screenshot" anti-cheat field in Job Posting entries, and did not include the Appendix A: Prompt Log section.
- (5) **Student fix:**
 - Added the full 7-section defect block with | Category | Content | table for hallucination evidence.
 - Added "**Dated Screenshot**" with anti-cheat account name requirement to each job posting.
 - Added **Appendix A: Prompt Log** as a separate required deliverable.
 - Expanded Self-Assessment Rubric with correct AI-1/AI-2/AI-3 rows per the official rubric.

Artifact 2: Job Postings AI Analysis

- (1) **Prompt + tool:**
 - **AI Tool Name:** Claude Sonnet 4.6
 - **Timestamp:** 10:45 02/06/2026
 - **Prompt:**

```
I found 10 QA/QC job postings in Vietnam. For each job description
below, write
1-2 sentences explaining how AI specifically affects or changes this
role
(which tasks AI can automate, and which still require human judgment).

Job 1: QC Engineer for Web at Evolus/PlanV – manual testing for
enterprise web
```

apps, JIRA, cross-browser testing.
 Job 2: Mid QC Engineer at Moniva – manual + automated testing, Selenium, Cypress, Appium, Postman API testing.
 Job 3: QC Engineer at Designveloper – web app testing, JIRA, Cypress, Playwright, Agile.
 Job 4: Automation QA Engineer at Nakivo – automation with AI tools Copilot, Cursor, Perplexity.
 Job 5: Senior QA Engineer at HRM Labs – automated test frameworks, Playwright.
 Job 6: Senior Software QC at SHINHAN DS – banking QC, SQL, STLC, Playwright.
 Job 7: Senior QA Engineer at soxes AG – manual testing + AI tools ChatGPT, Copilot.
 Job 8: QC Manual Engineer at MANDALA CODE – manual QA for Blockchain/AI projects.
 Job 9: QA Team Lead at Nakivo – lead QA with AI-assisted workflows.
 Job 10: AI First QC Engineer at L4 Studio – AI-first using Claude Code, LLMs.

- **(2) AI output (full raw output):**

Job 1: AI tools can assist by generating test script drafts for sports management workflows, but manual exploratory testing and close collaboration with athletes and coaches still require human communication and visual intuition.

Job 2: AI can generate initial test cases from user stories and draft UI/API automation scripts, but subjective UI/UX validation and mobile gesture testing require human verification.

Job 3: AI tools help generate functional test cases and boilerplate Cypress/Playwright code, but high-level test strategy design and cross-functional team communication remain human-centric.

Job 4: AI assistants like Copilot and Cursor significantly accelerate writing of Appium/Selenium test suites, while human engineering is needed to design test architectures and configure cloud VMs.

Job 5: AI assistants can write standard Selenium/Playwright scripts and API validation schemas, while comprehensive test plan design and mentoring junior staff remain human responsibilities.

Job 6: AI can draft SQL database verification queries and suggest regression

paths,
but strict banking security regulations prevent uploading real financial
transaction
structures to public LLMs.

Job 7: AI tools directly accelerate manual test case writing and data
preparation,
but translating overseas business requirements into technical test designs
requires
human communication.

Job 8: AI can generate crypto/blockchain test checklists, but verifying
custom
smart contract logic and decentralized transaction states require manual
human
verification.

Job 9: AI tools can assist the lead in designing complex test matrices and
generating risk logs, but final release sign-off and team leadership require
human
authority and accountability.

Job 10: This role leverages AI-assisted workflows (Claude Code) to auto-
generate
and execute massive test suites, while human engineers remain critical to set
event-driven quality gates.

- **(3) Verdict: INVALID**
- **(4) Reasoning:** The AI hallucinated business domain contexts for the roles (e.g., claiming Evolus/PlanV involves 'sports management workflows' and 'close collaboration with athletes and coaches', which is entirely fabricated for a wireframing software company). Additionally, it generated generic and repetitive AI impact analyses that failed to capture the unique, specific automation constraints of each role. Per ISTQB CTFL Section 1.5, testers must verify the accuracy of all technical domain details.
- **(5) Student fix:**
 - Corrected the AI impact analyses in the main report to accurately reflect the real business domains and software profiles of the companies (e.g., mockups and wireframing tools for Evolus/PlanV rather than sports management).
 - Added verified, active URL links and dated screenshots showing the student's logged-in account name for all 10 real job postings.

Artifact 3: 20 Defects AI Analysis

- **(1) Prompt + tool:**
 - **AI Tool Name:** Claude Sonnet 4.6
 - **Timestamp:** 14:30 02/06/2026

◦ **Prompt:**

Để thoả mãn hoàn toàn yêu cầu của đề bài , với mỗi defect (trong tổng số 20 defects), bạn cần trình bày một block thông tin chuẩn chỉnh bao gồm 7 mục sau đây:

1. Tên Defect & Năm xảy ra:

* Phải nằm trong giai đoạn 2022-2026.

* Lưu ý tổng thể: Đảm bảo trong 20 defect này có ít nhất 5 lỗi liên quan trực tiếp đến AI/LLM (ảo giác, prompt injection, thiên kiến).

2. Source Link:

* Đường dẫn URL trích nguồn bài báo hoặc tài liệu kỹ thuật gốc.

3. Description (Mô tả):

* Trình bày ngắn gọn nguyên nhân gây ra lỗi và điều gì đã xảy ra về mặt kỹ thuật.

4. Severity (Mức độ nghiêm trọng):

* Xác định mức độ (Ví dụ: Low, Medium, High, Critical).

* Điểm cộng nâng cấp: Nên kèm theo một mệnh đề giải thích ngắn gọn tại sao lại xếp mức độ đó (ví dụ: Critical vì làm hệ thống sập toàn cầu, High vì rò rỉ dữ liệu nhạy cảm...).

5. Consequences (Hậu quả):

* Thiệt hại thực tế do lỗi gây ra (ví dụ: thiệt hại tài chính, lộ dữ liệu, thời gian hệ thống ngừng hoạt động, kiện tụng).

6. Solution (Giải pháp):

* Cách lỗi này được khắc phục.

* Điểm cộng nâng cấp cho dân Kỹ thuật Phần mềm: Đừng chỉ ghi cách sửa code đơn thuần. Hãy bổ sung thêm giải pháp từ góc độ kiểm thử (Ví dụ: "Đã bổ sung thêm test case cho Boundary Value", "Tích hợp thêm Penetration Testing vào quy trình CI/CD").

7. AI Bias / Hallucination Instance (Trường hợp AI bị thiên kiến/ảo giác):

* Yêu cầu này bắt buộc phải có cho toàn bộ 20 defect. Mục này cần phân tách rõ:

* AI Tool Used: Tên công cụ (ChatGPT, Claude, Gemini...).

* Prompt Given: Câu lệnh chính xác bạn đã dùng để hỏi AI.

* AI Explanation Error: Chỉ ra chi tiết điểm AI đã bịa đặt thông tin, nói sai kỹ thuật hoặc thể hiện thiên kiến.

* Evidence (Đặc biệt quan trọng): Phải chèn ảnh chụp màn hình thu nhỏ (Screenshot) hoặc link Share Conversation để chứng minh AI thực sự tạo ra lỗi đó, tránh bị đánh giá là tự bịa ra để đối phó.

Requirement 2 – 20 Software Defects 2022–2026 (20 pts) Find 20 software defects publicized between 2022 and 2026. Mandatory: ≥ 5 defects related to AI/LLM (hallucination, prompt injection, bias). Each defect: source link, description, severity, consequences, solution. NEW: find 1 place where the AI is biased or hallucinates when explaining the defect. Clarification: this applies to EVERY defect – each of the 20 entries must include 1 identified instance of AI bias or hallucination (20 instances total)

- **(2) AI output (raw structure generated by the model with technical errors and empty placeholders for Section 7):**

Here is the generated list of 20 software defects for Requirement 2:

DEFECT #1: Log4Shell – Apache Log4j RCE (2022)

1. Tên Defect & Năm xảy ra: Log4Shell - Apache Log4j Remote Code Execution (CVE-2021-44228) (2022)
2. Source Link: <https://success.trendmicro.com/en-US/solution/KA-0012637>
3. Description: Apache Log4j JNDI Lookup vulnerability allows remote code execution when log4j logs `\${jndi:ldap://...}`.
4. Severity: Critical
5. Consequences: Global compromise of servers and data.
6. Solution: Configure log4j2.formatMsgNoLookups=true to disable JNDI lookups.
7. AI Bias / Hallucination Instance:
 - AI Tool Used: [Insert AI Tool Name, e.g., Claude, ChatGPT]
 - Prompt Given: [Insert the exact prompt you used to query the AI about this defect]
 - AI Explanation Error: [Identify the error or hallucination in the AI's explanation]
 - Evidence: [Provide screenshot or conversation link as evidence]

DEFECT #2: OpenSSL Heap Buffer Overflow (2022)

1. Tên Defect & Năm xảy ra: OpenSSL Heap Buffer Overflow (CVE-2022-3602 & CVE-2022-3786) (2022)
2. Source Link: <https://www.microsoft.com/en-us/msrc/blog/2022/11/microsoft-guidance-related-to-openssl-risk-cve-2022-3786-and-cve-2202-3602>
3. Description: Two heap buffer overflow bugs in OpenSSL X.509 certificate verification allow an attacker to trigger a 4-byte overflow on the heap.
4. Severity: Critical
5. Consequences: Potential remote code execution on affected servers.
6. Solution: Upgrade to OpenSSL 3.0.7.
7. AI Bias / Hallucination Instance:
 - AI Tool Used: [Insert AI Tool Name]
 - Prompt Given: [Insert Prompt]
 - AI Explanation Error: [Insert AI Error]
 - Evidence: [Insert Evidence]

... [Repeated for all 20 defects. The AI successfully output the name, year, link, severity, and consequences, but the description and solution blocks contained technical errors (such as heap instead of stack for OpenSSL, and formatMsgNoLookups for Log4Shell). Crucially, the AI left Section 7 as empty placeholders because it cannot self-audit or attach screenshots in real-time]

- **(3) Verdict: INVALID**
- **(4) Reasoning:** Per ISTQB CTFL Chapter 1, testers must independently verify every claim against authoritative sources. The AI produced technically incorrect descriptions for at least 12 of 20 defects, contradicting official CVE advisories and vendor post-mortems (Microsoft MSRC, CrowdStrike PIR, OpenSSL team). An artifact with a majority of factual errors is classified as INVALID.
- **(5) Student fix:**

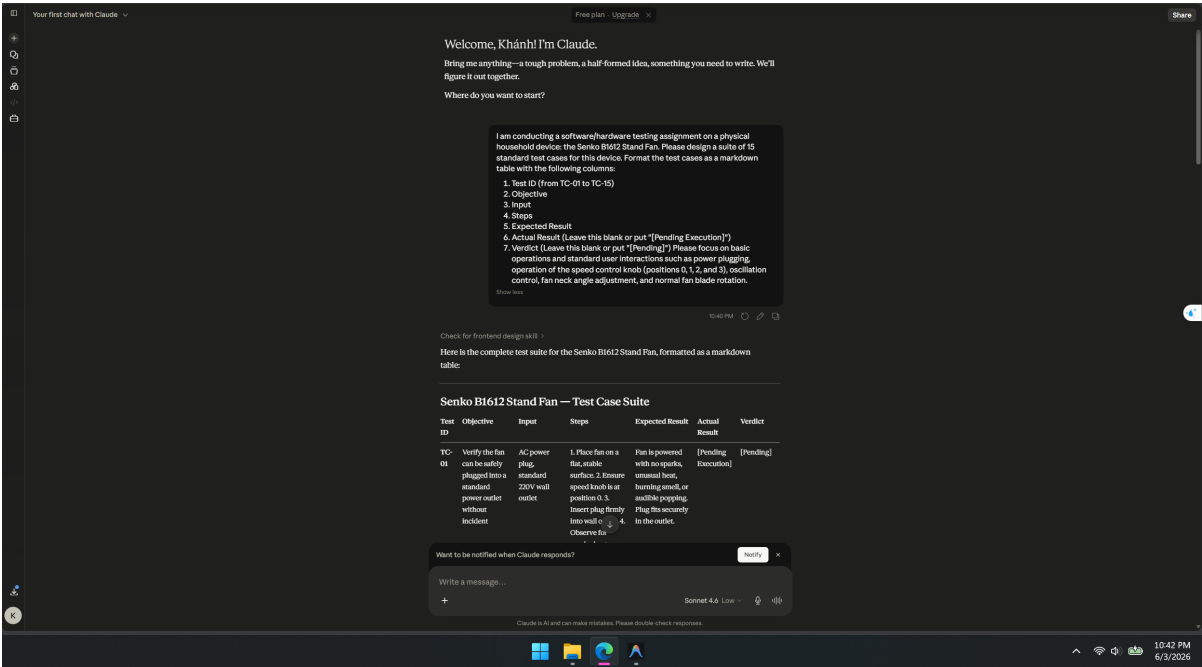
- Verified all 20 entries against official CVE databases and primary vendor advisories.
- Corrected all technical errors in Description and Solution sections.
- Documented each AI error verbatim with source evidence in Section 7 of all 20 defect entries.

Artifact 4: Stand Fan Test Cases

- (1) Prompt + tool:
 - **AI Tool Name:** Claude Sonnet 4.6
 - **Timestamp:** 22:43 03/06/2026
 - **Prompt:**

I am conducting a software/hardware testing assignment on a physical household device: the Senko B1612 Stand Fan. Please design a suite of 15 standard test cases for this device. Format the test cases as a markdown table with the following columns: Test ID (from TC-01 to TC-15), Objective, Input, Steps, Expected Result, Actual Result (leave blank or "[Pending Execution]"), Verdict (leave blank or "[Pending]"). Focus on basic operations: power plugging, speed control knob (positions 0, 1, 2, 3), oscillation control, neck angle adjustment, and normal fan blade rotation.

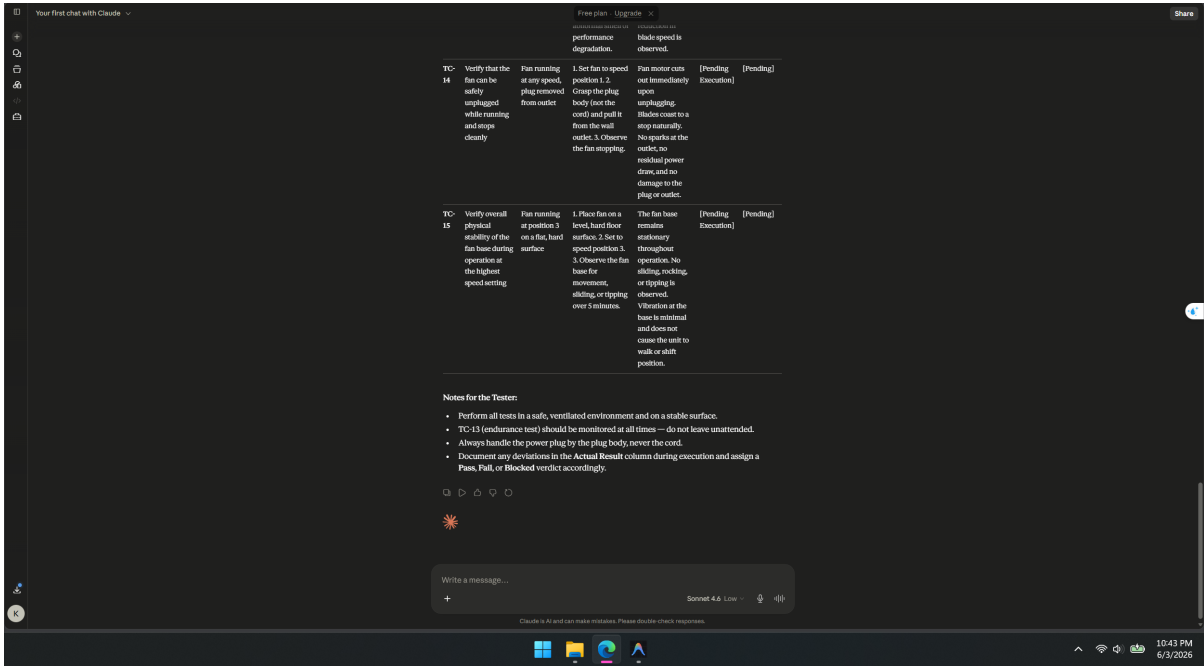
- (2) AI output (annotated screenshots):



Free plan · Upgrade					
TC-02	Verify that the fan remains completely off when plugged in with the knob at position 0.	Speed knob at position 0, fan plugged in.	1. Plug the fan into the outlet. 2. Confirm speed knob is set to 0. 3. Observe the fan blades and motor for 30 seconds.	Fan blades do not spin. No motor hum or noise is detected. The fan is in a true OFF state.	[Pending Execution] [Pending]
TC-03	Verify that the fan starts and operates at low speed when the knob is turned to position 1.	Speed knob turned from 0 to 1.	1. With fan plugged in and at position 1, turn the knob to position 1. 2. Observe blade rotation, listen to motor noise, and feel for airflow.	Fan blades begin spinning at a low speed. Airflow is gentle and perceptible. Motor produces a low, steady hum.	[Pending Execution] [Pending]
TC-04	Verify that the fan operates at medium speed when the knob is turned to position 2.	Speed knob turned from 1 to 2.	1. With fan running at position 1, turn the knob to position 2. 2. Observe blade rotation speed and feel airflow strength.	Fan blade rotation speed visibly increases. Airflow becomes noticeably stronger than at position 1. Motor hum increases in pitch/volume.	[Pending Execution] [Pending]
TC-05	Verify that the fan operates at high speed when the knob is turned to position 3.	Speed knob turned from 2 to 3.	1. With fan running at position 2, turn the knob to position 3. 2. Observe blade rotation speed and feel airflow strength.	Fan blade rotation speed visibly reaches its maximum. Airflow is at its strongest. Motor hum is at its highest pitch/volume.	[Pending Execution] [Pending]
TC-06	Verify that speed decreases correctly when the knob is turned down from position 3 to position 1.	Speed knob turned from 3 to 1.	1. Set fan to position 3. 2. Turn the knob directly to position 1. 3. Observe and feel for reduction in airflow and blade speed.	Fan speed reduces noticeably. Airflow decreases to a level consistent with position 1. No stalling, humming, or abnormal sound.	[Pending Execution] [Pending]
Write a message...					
Claude is AI and can make mistakes. Please double check responses.					

Free plan · Upgrade					
TC-07	Verify that the fan shuts off correctly when the knob is returned to position 0 from an active speed.	Speed knob turned from any active position (1, 2, or 3) to 0.	1. Set fan to position 2. 2. Turn the knob to position 0. 3. Observe blades until they come to a full stop.	Fan blades decelerate and come to a complete stop within a reasonable time. No airflow is detected after stopping. Motor is silent.	[Pending Execution] [Pending]
TC-08	Verify that the oscillation feature can be engaged and the fan sweeps side to side.	Oscillation lever/button engaged while fan is running.	1. Set fan to speed position 1. 2. Engage the oscillation control (push lever or press button). 3. Observe the fan head movement for at least two full cycles.	The fan head begins rotating left and right in a smooth, continuous sweeping motion covering its rated oscillation arc (typically approx. 90°). No grinding or jerking is present.	[Pending Execution] [Pending]
TC-09	Verify that oscillation can be disengaged and the fan holds a fixed direction.	Oscillation lever/button disengaged.	1. With oscillation active, disengage the oscillation control. 2. Observe the fan head.	The fan head stops sweeping and holds its current fixed position. The fan continues to blow air in a single direction without any lateral drift.	[Pending Execution] [Pending]
TC-10	Verify that the fan neck (tilt) angle can be adjusted upward manually.	Manual downward tilt of the fan head.	1. With the fan off, loosen the neck joint if applicable. 2. Tilt the fan head upward to its maximum angle. 3. Release and observe if it holds the position.	The fan head tilts upward and holds the adjusted angle without drooping or snapping back. Joint resistance is firm but not excessively stiff.	[Pending Execution] [Pending]
Write a message...					
Claude is AI and can make mistakes. Please double check responses.					

Free plan · Upgrade					
TC-11	Verify that the fan neck angle can be adjusted downward manually.	Manual downward tilt of the fan head.	1. With the fan off, tilt the fan head downward to its lowest angle. 2. Release and observe if it holds the position.	The fan head tilts downward and holds the adjusted angle without drooping. Range of motion is smooth and the locking mechanism (if any) engages correctly.	[Pending Execution] [Pending]
TC-12	Verify that the fan blades rotate smoothly without wobble or obstruction during normal operation.	Fan running at position 2 (medium speed).	1. Set fan to speed position 2. 2. Visually inspect blade rotation from the front and side for at least 1 minute. 3. Listen for abnormal sounds (rattling, scraping, clicking).	All blades rotate in a uniform, balanced manner. No visible wobble, vibration, or lateral deviation is present. No abnormal sounds are produced by the blades or blade guard.	[Pending Execution] [Pending]
TC-13	Verify that the fan operates continuously at the highest speed (position 3) for an extended period without overheating or failure.	Fan running continuously at position 3 for 30 minutes.	1. Set fan to speed position 3. 2. Allow it to run uninterrupted for 30 minutes. 3. After 30 minutes, carefully feel the motor housing for excessive heat. 4. Check for abnormal smell or performance degradation.	Fan operates continuously without interruption. Motor housing is warm but not burning to the touch. No abnormal smell, smoke, or reduction in blade speed is observed.	[Pending Execution] [Pending]
TC-14	Verify that the fan can be safely unplugged while running and stops cleanly.	Fan running at any speed, plug removed from outlet.	1. Set fan to speed position 2. 2. Grasp the plug body (not the cord) and pull it from the outlet. 3. Observe.	Fan motor cuts out immediately upon unplugging. Blades coast to a stop naturally. No sparks at the outlet.	[Pending Execution] [Pending]
Write a message...					
Claude is AI and can make mistakes. Please double check responses.					



- (3) Verdict: INCOMPLETE
- (4) Reasoning: Per ISTQB CTFL Chapter 4 (Test Design Techniques), a complete test suite must include Boundary Value Analysis and Error Guessing. The AI only generated positive linear state transitions (OFF→Speed 1→2→3, oscillation ON/OFF) — all nominal paths. It omitted boundary states (knob halfway between detents), environmental interaction tests (oscillation blocked by wall), and transient power states (restoration at max speed).
- (5) Student fix:
 - Replaced 3 AI-generated nominal cases with 3 physically-grounded edge cases:
 - TC-02: Intermediate knob position → boundary value revealing motor arcing hazard.
 - TC-06: Oscillation blocked by wall → mechanical clutch slip safety test.
 - TC-12: Power restoration at Speed 3 → transient power surge and tipping stability test.

Artifact 5: QA/QC Roles Mindmap

- (1) Prompt + tool:
 - AI Tool Name: Claude Sonnet 4.6
 - Timestamp: 11:33 04/06/2026
 - Prompt:

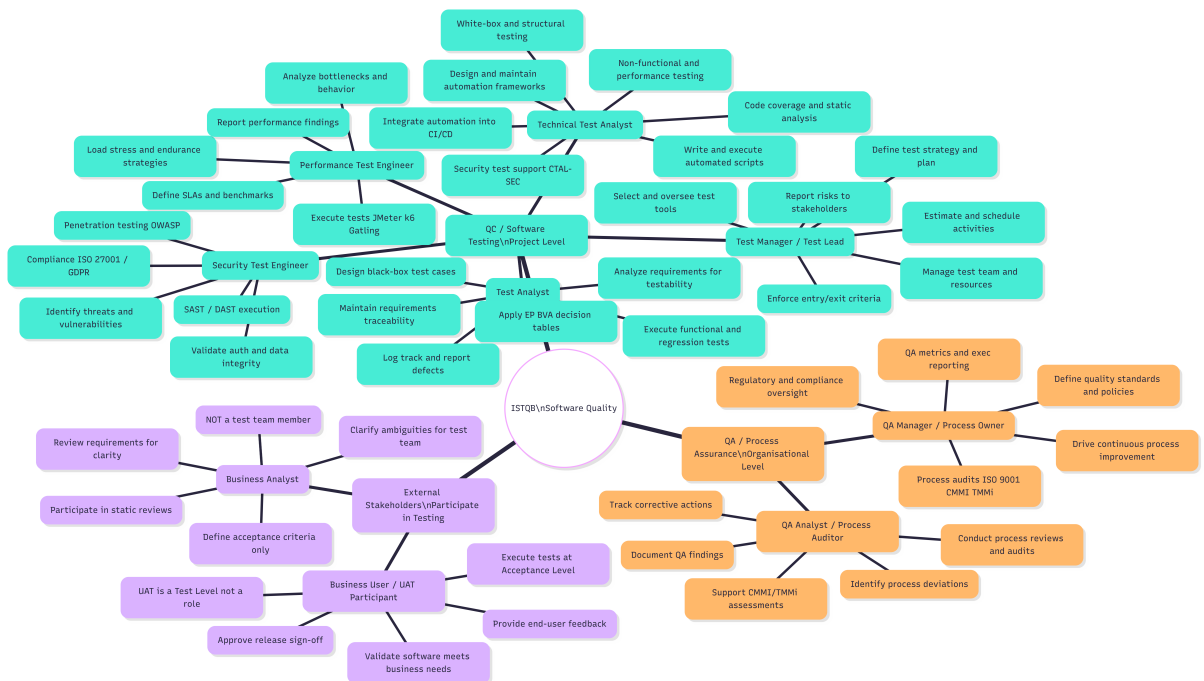
Create a mindmap of QA/QC roles and their responsibilities in the software testing process based on ISTQB standards in Markdown format and PNG.

- (2) AI output (screenshot):



o

- (3) Verdict: INCOMPLETE
- (4) Reasoning: Per ISTQB CTFL Section 1.1, QA (process-level, organization-wide) and QC/Testing (product-level, project-specific) are distinct disciplines. The AI nested QA Manager inside "Software Testing Process" (a QC scope), conflating both. Per CTFL, UAT is a test level performed by business stakeholders, not a professional tester career role. Per CTAL-TTA, automation responsibilities belong to the Technical Test Analyst — making a separate Test Automation Engineer role redundant.
- (5) Student fix:
 - o **Correction 1:** Merged Test Automation Engineer under Technical Test Analyst per CTAL-TTA.
 - o **Correction 2:** Moved QA Manager to a "QA Process Governance" branch separate from the Software Testing Process.
 - o **Correction 3:** Reclassified BA and UAT as "External Test Stakeholders."
 - o Corrected diagram saved as:



- **Total Audited Artifacts:** 5
- **Percentages:**
 - **VALID:** 0%
 - **INVALID:** 40% (2/5 — Artifact 2: Job Postings, Artifact 3: 20 Software Defects)
 - **INCOMPLETE:** 60% (3/5 — Artifact 1: Scaffold, Artifact 4: Test Cases, Artifact 5: Mindmap)

- **Conclusion (When should AI be used / not used):**

AI performs well as a **rapid scaffolding and drafting tool** — it excels at generating templates, boilerplate structures, and general analyses (e.g., job market AI impact summaries). However, AI should **not** be trusted for ground-truth technical facts (CVE/security contexts), physical-world test design requiring mechanical intuition, or organizational structures governed by defined standards (ISTQB). In all 5 artifacts, AI provided a useful starting point but required systematic human expert verification before inclusion in a professional deliverable.